WORKING PAPER · DRAFT OF 17 OCTOBER 2026 · NOT PEER REVIEWED

Video Jev: one calibrated head for typed questions about video and sound

Ask a video a typed question: yes or no, pick one, score it, or "when?". Get back a probability for every possible answer. Many questions in one pass.

Backbone: Qwen3-Omni-30B-A3B (Apache 2.0) Compute: Modal, 1 × H200 Spend to date: $250 billed

Code: multimodal-jev

SUMMARY

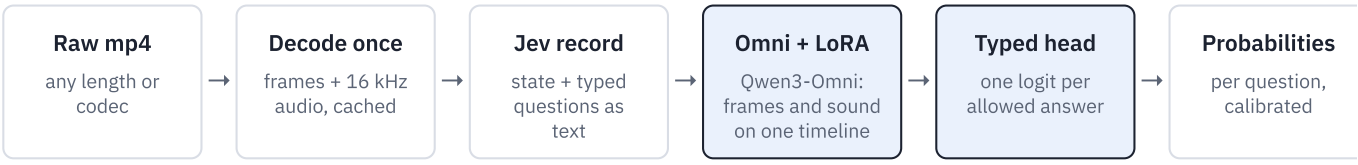
- **What it is.** We add a small trained head to an open audio-visual model, Qwen3-Omni. The head reads questions written in Jev form and returns a calibrated probability for every allowed answer. One model, one forward pass, many questions.
- **What works.** On five tasks, training lifts the same model far above its untrained self. It matches specialist systems on two of them.
- **Engagement (SnapUGC):** 0.709 on held-out test, level with the 2025 challenge winner's 4-model ensemble (0.710, different split).
- **AI-video quality (VideoFeedback):** 0.775 average Spearman on the official test split, against VideoScore's 0.771. Ahead on 3 of 5 questions, from a general recipe with no task tuning.
- **"When" questions:** a new `moment` type, with no new parts, finds the funny moment in FunQA 81% of the time, against 53% for the best blind guess.
- **Sound:** on "which sound is this?", audio adds 6.4 points and cuts calibration error by 64%.
- **What does not work yet.** Text-query moments on QVHighlights: a small run is behind the untrained model on spans. Real-or-AI ties a CLIP baseline on old generators.
- **The ask.** About $150 trains one joint model on all tasks, tests it on question wordings it never saw, scores it once on every test split, and releases it on Hugging Face for non-commercial use.

0.709	0.775	+27 pts	0.02 to 0.06
Engagement, test	AI-video quality, test	Funny moment HIT@1	Calibration error (ECE)
2025 winner: 0.710	VideoScore: 0.771	over best blind guess	untrained: 0.18 to 0.56

1. What we are building

Jev is a format for decisions. A record holds some *state* (context) and a set of typed *questions*. Each question lists its allowed answers. A Jev model returns a probability for each answer, and code acts on those numbers.

Clef, the existing Jev model, reads text and images. **Video Jev** reads raw video with its sound. The goal is a general head: write a new question in Jev form, and the model answers it with a probability you can trust.



The pipeline. Only the LoRA adapters and the head are trained. The rest of Qwen3-Omni is frozen.

Four question types

TYPE	ANSWER	EXAMPLE QUESTIONS	SHOWN ON
`noul`	P(yes), P(no)	"Does this look AI-generated?" "Did the crowd cheer?"	GenVideo, VideoFeedback
`choice`	P per named option	"Which sound is this: laughing, giggling, cheering, speech...?"	VGGSound
`score`	P per ordered level	"How good is the visual quality, 1 to 4?" "What share of viewers keep watching?"	SnapUGC, VideoFeedback
`moment` new	P per second	"When is the funny moment?" "When does the goal go in?"	FunQA, QVHighlights

An example record (illustrative)

```
{"state": {"title": "my dog vs the hoover"},
 "questions": {
  "keeps_watching": {"type": "score",
     "levels": ["0-10%", ..., "90-100%"]},
  "got_a_laugh": {"type": "noul",
     "instructions": "Did this get a laugh?"},
  "funniest": {"type": "moment",
     "instructions": "When is the funniest
moment?"},
  "sound": {"type": "choice", "options":
     ["laughing", "cheering", "speech", ...]}}}
```

```
keeps_watching  60-70%: 0.31  70-80%: 0.27 ...
got_a_laugh     yes: 0.84   no: 0.16
funniest        5s: 0.26  6s: 0.30  7s: 0.12 ...
sound           laughing: 0.71  speech: 0.18 ...

one forward pass, about 0.2 s on one H200
```

Illustrative only. This shows the shape of the input and output, not a real model output. Each question here has been trained separately; G4 trains them together.

Why probabilities, and why calibration

A model that says "yes" is hard to build on. A model that says "yes, 84%" lets code set thresholds, rank clips, and send unsure cases to a person. That only works if 84% means right about 84% of the time. We measure this with **Brier score** (squared error of the probabilities) and **ECE** (expected calibration error: the gap between stated confidence and actual accuracy). Most video models report accuracy only.

2. How it works

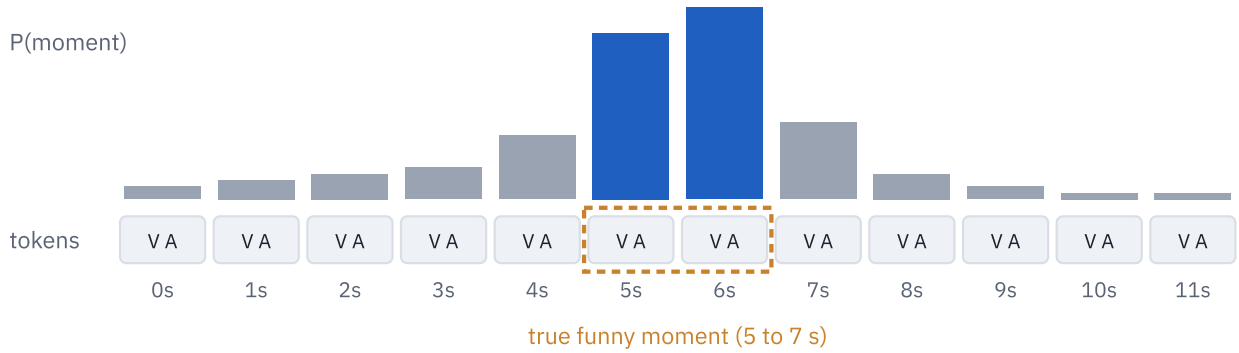
The head

The answer options are written in the prompt. The head reads the model's hidden states over each option's tokens and gives that option one logit. A softmax over a question's options gives its probabilities. All questions in a record are answered in one forward pass, so ten questions cost about the same as one.

The `moment` trick

Qwen3-Omni lays video and audio tokens on one timeline, in 2-frame blocks. For a "when" question, each time bin *is* an option. The option's tokens are that bin's own video and audio

tokens. The same head scores every bin, so the answer is a probability for every second. This needs no detector and no generated timestamps, and it can report the top few moments and how sure it is.

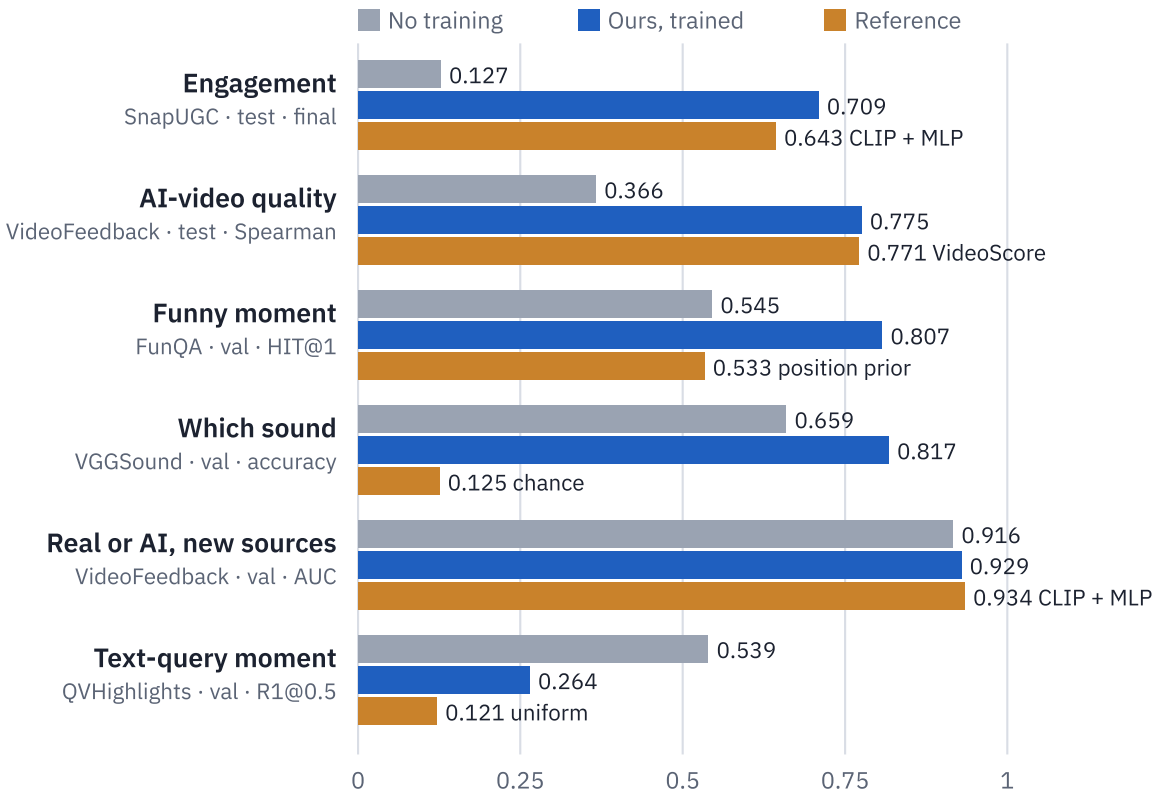


Each block of video (V) and audio (A) tokens is one answer option. The head gives a probability per second (bars). Illustrative shape; real FunQA results are in section 3.

Training

- **Trainable:** LoRA (rank 64) on the language model's attention, plus the head. About 0.2% of parameters.
- **Loss:** cross-entropy with soft targets, plus Brier score, plus an ordinal distance (EMD) for score and moment questions. Calibration comes from this loss; no step after training is needed.
- **Recipe:** LoRA learning rate 3e-5, head 1e-4, 1 epoch, cosine schedule, 8-clip gradient accumulation. Found on SnapUGC and reused unchanged on every new task.
- **Splits:** official test sets where they exist, scored once per final model. Choices are made on val only.

3. Results



Every task on one 0 to 1 scale, each with its own metric. "No training" is the same Qwen3-Omni asked the question as text, reading the probability of each answer word (SnapUGC uses untrained Clef-flash). SnapUGC and VideoFeedback on test; the rest on val. QVHighlights is a small run on a third of the training data.

3.1 Engagement: SnapUGC (test, 1,736 clips, scored once)

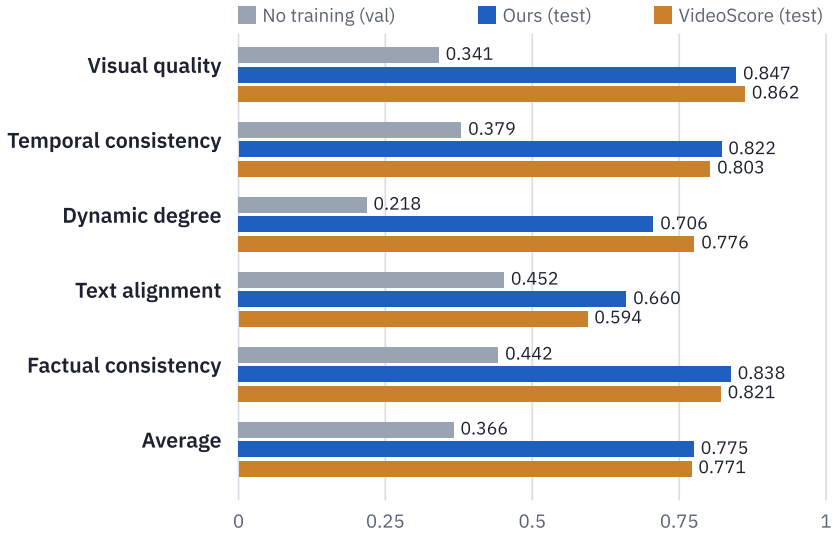
MODEL	FINAL	BRIER	ECE
Omni + typed head	**0.709**	**0.818**	**0.028**
CLIP frames + MLP	0.643	0.843	0.034
Text only (XGBoost)	0.347	0.884	0.029
Clef-flash, untrained	0.127	0.912	0.064
2025 winner, 4-model ensemble*	0.710	-	-

Final = mean of Spearman and Pearson. *Official test split; ours is a different held-out split, so "level with", not "beats". Gap to CLIP: +0.065, 95% CI 0.041 to 0.088.

The question: what share of viewers keep watching past 5 seconds, as a 10-level score. 30,631 training clips, video + audio + title.

One model matches a four-model ensemble. Its probabilities are calibrated with no extra step.

3.2 AI-video quality: VideoFeedback



Five score questions per clip, all in one pass. 35,041 training clips, 4,000 steps.

- **Official test split, 760 clips, scored once:** 0.775 against VideoScore's 0.771.
- Ahead on temporal consistency, text alignment and factual consistency.
- Behind on visual quality (0.847 against 0.862) and dynamic degree, the amount of motion (0.706 against 0.776).
- Calibration error 0.043 overall. VideoScore gives no probabilities.
- Test (0.775) came in above val (0.755). One seed.

3.3 "When" questions: FunQA and QVHighlights

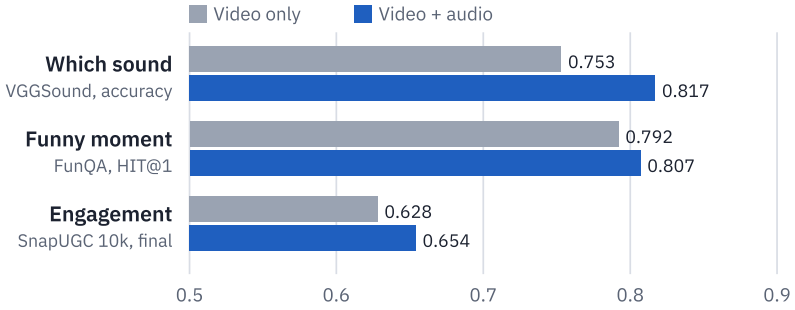
FUNQA (400 VAL VIDEOS)	HIT@1	R1@0.5	MIOU
Uniform guess	0.435	0.320	0.380
Position prior	0.533	0.135	0.190
Untrained Omni (text answer)	0.545	0.295	0.350
Ours, video + audio	**0.807**	**0.580**	**0.538**

HIT@1: the most likely second is inside the true moment. R1@0.5: the predicted span overlaps the true one by IoU 0.5 or more. Funny, creative and magic moments.

QVHIGHLIGHTS (356 VAL)	HIT@1	R1@0.5	MIOU
Uniform guess	0.089	0.121	0.263
Untrained Omni (text answer)	0.195	0.539	0.520
Ours, small run (2k queries)	0.213	0.264	0.264

Text queries like "a woman walks her dog". The untrained model was likely trained on this kind of data already. Ours finds the right second a little more often but draws spans that are too wide.

3.4 Sound: does audio help?



Same model and recipe, audio on or off. Axis starts at 0.5.

- **VGGSound** (8 people-sound classes: laughing, giggling, baby laughter, cheering, crowd, woman, man and child speech): audio adds 6.4 points, and ECE falls from 0.058 to 0.021. This is the clean audio result.

- **FunQA**: no clear gain. Its moments are visual: pranks, tricks, magic.
- **SnapUGC**: a small gain inside the noise.
- So the model uses sound when the answer is in the sound.

3.5 Real or AI

On GenVideo (10 generators from 2023 to 2024) and across to VideoFeedback's sources, our model ties a CLIP baseline (AUC 0.929 against 0.934). These older generators are easy to spot from single frames. A useful test needs modern generators (Sora or Veo class), and no open labelled set of them exists yet.

4. What we tried, and what we learned

DATE	WHAT WE TRIED	WHAT HAPPENED	LESSON
7 Oct	Untrained Clef-flash on SnapUGC	0.127	Prompting is not enough; training is the whole game.
8 Oct	Pilot, 10k clips, Qwen3-Omni + head	0.654 val	The design works; CLIP + MLP was level, so more data was needed.
8 Oct	Full run at LoRA rate 1e-4	collapsed to 0.19 after 400+ steps	Short sweeps cannot see this failure. 3e-5 is stable.
8 Oct	Head in bf16	NaN around step 720, twice	Head now runs in fp32; bad steps are skipped and counted.
9 Oct	Full run at 3e-5, 30k clips; test once	0.709 test	One model is level with a 4-model ensemble.
9 Oct	Omni + CLIP ensemble	+0.001 on test	Ensembling stops helping as the model gets stronger.
10 Oct	Same recipe on VideoFeedback	0.653 after 2k clips, 0.775 test at full	The recipe transfers with no tuning.
10 Oct	Real or AI on GenVideo	0.999 shortcut	Real clips were all 10 s, fakes shorter. We now hide length, size and shape. Probe every dataset first.
10 Oct	New `moment` type on FunQA	0.807 HIT@1	"When" questions work with no change to the head.
10 Oct	VGGSound with and without audio	+6.4 pts	First clean audio gain.
10 Oct	QVHighlights, small run	behind on spans	Long clips (150 s) and a strong untrained model: needs full data and sharper spans.
10 Oct	Untrained Omni on every task	0.37 to 0.92	Training doubles VideoFeedback agreement; untrained probabilities are badly calibrated.

What surprised us

- **The recipe did not need retuning.** Settings found on engagement worked on quality, moments, sound and real-or-AI.
- **Calibration is close to free.** ECE stayed between 0.02 and 0.06 on every trained task, with no temperature scaling. Untrained, it was 0.18 to 0.56.
- **Datasets have traps.** One real-or-AI set could be solved from clip length alone. A baseline at 0.999 is a warning, not a result.

5. Where we are now

PHASE	TASK	STATUS	RESULT
SnapUGC	Engagement	done, parked	0.709 test; 3 seeds later
G0	AI-video quality	done	0.775 test, VideoScore 0.771
G1	Real or AI	done, tie	Ties CLIP; old generators too easy
G2	Text-query moments	small run only	Behind untrained model on spans
G3	Funny moments, sound	done	0.807 HIT@1; audio +6.4 pts
G4	One joint model, unseen questions	not started	The main result. Needs the $150.
G5	Long video	not started	After release

Spend so far: $250 billed

ITEM	APPROX. COST
SnapUGC: data, pilot, failed runs, full run, test	$157
Generalisation G0 to G3, zero-shot baselines	$62
Storage (volumes: videos, caches, checkpoints)	$32
Billed total (after $30 credit and free storage)	**$250**

About $50 of that was runs we learned from and then replaced: the learning-rate collapse, the NaN incident and the GenVideo shortcut. Those lessons are now built into the code, which is why later phases cost $5 to $25 each.

Storage keeps billing monthly. We can delete raw downloads we no longer need.

6. What $150 gets you

One released model that answers all of these question types, with honest test numbers and a clear statement of what it cannot do.

#	STEP	COST	WHAT YOU GET	GATE (DECIDED BEFORE RESULTS)
1	Score G0 on the official VideoFeedback test split, once	done	**0.775 against VideoScore's 0.771** (paid from the raised limit, 17 Oct)	-
2	Small joint test: all tasks, 10% of data	$10	Proof the tasks train together without breaking	No task falls more than 0.05 below its single-task run.
3	**G4 joint model**: VideoFeedback, FunQA, QVHighlights (all of it), VGGSound, real-or-AI, with questions reworded in many ways	$70 to $90	**The release model.** One set of weights for score, yes/no, choice and moment questions.	Within 0.02 of single-task models on VideoFeedback and FunQA.
4	Unseen-question test	in 3	Evidence the head is general: questions it never trained on	Beats the untrained model on held-out questions.
5	Score every test split once; untrained baseline on the same clips	$18	The table on the model card	-
6	Buffer for one rerun	$25	Every phase so far needed one	Unused buffer is not spent.

7	Release on Hugging Face	$0	LoRA + head (a few hundred MB), loader script, Jev record examples, model card	Licence check of every dataset first.
	Total	**$123 to $143**	**Leaves about $7 to $27 of the $150 spare**	

How we test "general"

- **New wordings.** Each question is trained with several paraphrases, and tested with paraphrases it never saw.
- **New question families.** Hold one out completely. For example, train on funny and magic moments, then ask for the *creative* moment. Or train four VideoFeedback questions, then ask the fifth.
- If the model beats its untrained self on these, the head generalises beyond its training questions. That is what makes it a general Jev head, not five task models in one file.

> WHAT $150 DOES NOT BUY
>
> - More seeds (about $140 for two more), so headline numbers come from one run each.
> - A fix for QVHighlights beyond the full-data run. If spans stay wide, we report it as a limit.
> - Detection of modern AI video, sports goal timing or long videos. These need data or work we do not have yet.
> - Commercial use. FunQA is CC BY-NC-SA, so the release is non-commercial.

What it costs to use

INPUT (ONE H200, BATCH 1, MEASURED)	FORWARD PASS	PER 1,000 CLIPS
1.5 s clip, video only (real or AI)	133 ms	~$0.18
10 s clip, video + audio (VGGSound)	206 ms	~$0.27
16 s clip, video + audio + title (SnapUGC)	290 ms	~$0.39
Up to 60 s, video + audio (FunQA)	230 ms	~$0.31
150 s, video + audio at 1 fps (QVHighlights)	1,140 ms	~$1.52

At about $4.80 per GPU hour; decoding not included. The bf16 model needs an 80 GB GPU. A 4-bit version (about 17 GB, untested) could fit a 24 GB consumer card. Each forward pass answers every question in the record.

7. Use cases

Ready means the question type is trained and measured. Partly means one part is shown and the rest needs data. Each use case is a Jev record: you write the questions and get probabilities back.

Clip finder for creators

ready

"When is the funniest moment?" "Will viewers keep watching?" Find the best seconds of a long video and rank cuts by likely engagement. This follows directly from the clipping-for-distribution work.

Reviewer for AI video

ready

"Visual quality 1 to 4?" "Does it match the prompt?" "Does the physics hold?" Score every generation, drop bad takes, and give a generator a reward signal (the generate-then-review loop).

Sound-aware tagging

ready

"Is that laughing, cheering or speech?" "Did this get a laugh?" Tag clips by what you hear, not just what you see, with a probability per tag.

Triage with a human in the loop

ready

Because the probabilities are calibrated, auto-accept above 95%, auto-reject below 5%, and send the rest to a person. A plain classifier cannot set these thresholds safely.

Dataset labelling for research

ready

Label a large video set with several typed questions in one pass each. Use the uncertain clips for human labelling next (active learning).

Find a moment by text

partly

"When does the woman open the door?" Search inside long videos. The mechanism works; quality on QVHighlights is not there yet.

Sports and live events

partly

"When does the goal go in?" "Did the crowd cheer?" The moment and sound parts both work. It needs labelled match data (SoccerNet needs a signed agreement).

Spotting modern AI video

not yet

"Was this AI-generated?" Works on old generators, but so does CLIP. Modern generators need a new labelled set.

8. Related work

Each part has been done before. We found no work that combines them. We do not claim novelty; we want a good general head, compared fairly with what exists.

WORK	WHAT IT DOES	HOW WE DIFFER
Q-Align (2023)	Quality scores from probabilities over rating words	Uses the language model's vocabulary; scores only
VideoScore (2024)	AI-video quality on 5 fixed questions; source of VideoFeedback	Our direct baseline. No audio, fixed questions
DPC-VQA (2026)	Logits over ordered quality labels, plus calibration	Closest to our score head; one task only
UVE (2025)	Yes/no probabilities to score AI video	No learned head, no "when"
Engagement with LMMs (2025)	SnapUGC engagement; audio helps	Numbers as generated text; no calibration
Moment-DETR, UniVTG (2021, 2023)	Dedicated moment-finding models	Separate heads per task, not typed questions
TimeChat, VTimeLLM (2023)	Video models that write timestamps as text	One answer, no probability over time
"Your VLM Already Knows When" (2026)	Finds moments by yes/no readout, no training	Untrained; we train with soft targets
ReVisionLLM (2024)	Reports calibration for moment confidence	Grounding only
v-HUB (2025)	Video humour benchmark with audio	A benchmark, not a model

First-pass search from paper summaries; full reading is needed before submission. Details in docs/related_work.md.

9. Limitations

- **One seed per result.** We do not yet know the run-to-run spread.
- **Val, not test, for three tasks.** SnapUGC and VideoFeedback are scored on held-out test. Step 5 of the $150 covers the rest.
- **SnapUGC split differs from the challenge.** The official test labels are withheld.
- **Only 32% of SnapUGC videos are still online,** and the deleted ones skew to low engagement.
- **Short clips.** Up to 16 s for engagement and 150 s for QVHighlights. Long video is untested.
- **Real-or-AI uses old generators** and ties CLIP.
- **Licences:** research and non-commercial use only. We release weights, never the videos.

10. Datasets and licences

DATASET	TASK	TRAIN CLIPS	LICENCE
SnapUGC (VQualA 2025)	Engagement score	30,631	Challenge terms (research)
VideoFeedback (TIGER-Lab)	5 quality scores, real or AI	35,041	Apache 2.0 (to confirm)
GenVideo	Real or AI	3,652	To confirm before release
FunQA	Funny, creative, magic moments	3,072	CC BY-NC-SA 4.0
QVHighlights	Text-query moments	6,862	CC BY-NC-SA 4.0 (to confirm)
VGGSound	Which sound	5,506	CC BY 4.0

No private data is used. Full run logs: results/README.md. Runs: W&B project sp1d5r/vjev.